

XAUUSD 4H

TRADING DECISION SUPPORT PLATFORM

Complete Product Plan, Quantitative Trading Logic, Fundamental Analysis Framework, Risk Engine, System Architecture and Master AI Build Prompt

Primary market	XAUUSD (Gold vs USD)
Primary timeframe	4-hour (H4), with D1 context and H1 confirmation
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Design intent

This system is designed as a risk-first decision-support platform. It may output BUY, SELL, WAIT FOR CONFIRMATION, or NO TRADE. It must never force a trade merely because a daily profit target has not been reached.

Prepared as a build-ready blueprint for use with an AI coding agent, engineering team, or individual developer.

Document purpose and usage

This specification consolidates the complete application concept into one build-ready source of truth. It covers product behavior, market data, technical and fundamental analysis, position sizing, risk controls, paper trading, backtesting, performance analytics, architecture, database design, APIs, testing, deployment and a master implementation prompt.

Important

The USD 10/day figure is a user-defined progress target, not a promised return and not an input to position sizing. The sizing sequence is always: equity -> risk limit -> technical stop distance -> broker symbol specification -> safe volume -> estimated outcome.

Trading risk

Leveraged trading can produce rapid losses. Live execution should remain disabled until the strategy has passed paper trading, out-of-sample testing and jurisdiction/broker review. The final decision to place any trade remains with the user.

Guiding principles

- Capital preservation comes before the daily profit target.
- No trade is a valid and often preferred outcome.
- Market bias, trade setup and risk permission are separate outputs.
- All confirmed H4 signals use closed candles; intrabar observations are explicitly labeled unconfirmed.
- Fundamental drivers are contextual, not deterministic. Relationships between gold, the USD, yields and risk can weaken or reverse in different regimes.
- The LLM may explain structured evidence, but deterministic engines own calculations, gating and risk controls.
- No current price, macro statistic or news item may be invented by the LLM.

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1. Executive summary

The application is a web-based XAUUSD decision-support and paper-trading platform centered on the H4 timeframe. It continuously converts market data, macro/fundamental context, economic-event risk and account constraints into an explainable action state. The system should not be optimized to produce frequent signals; it should be optimized to reject weak or unsafe trades.

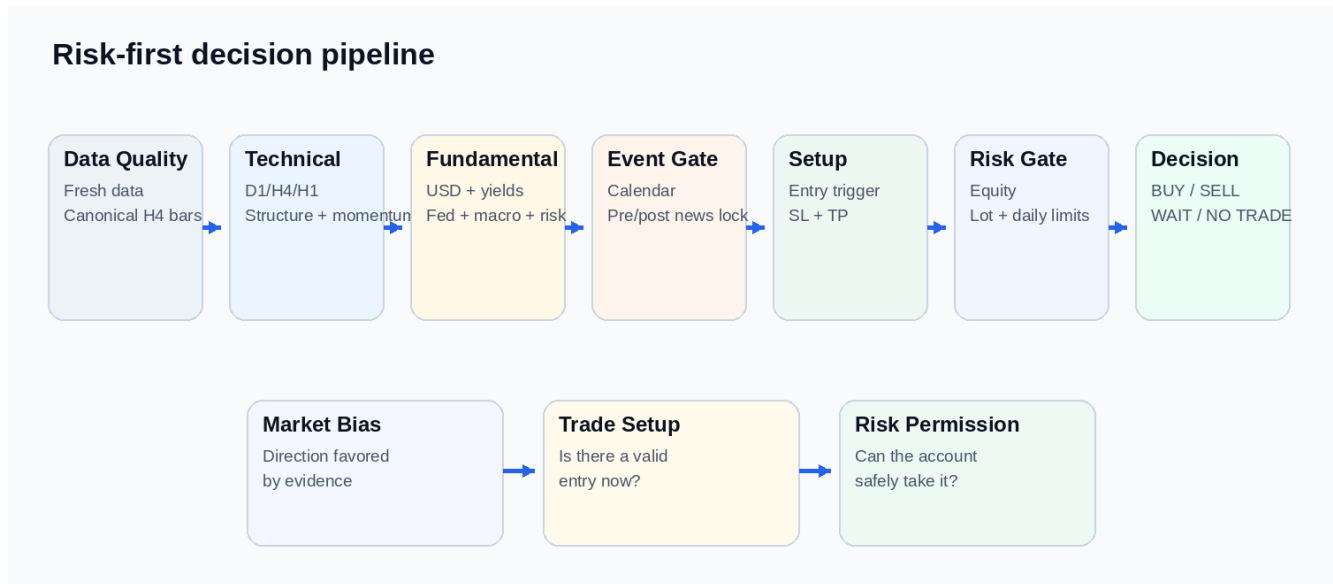


Figure 1. The final action is produced only after data, setup and account-safety gates pass.

Core output model

Output	Meaning	Example
Market Bias	Directional evidence independent of entry timing	BULLISH
Trade Setup	Whether a valid entry exists now	WAITING FOR PULLBACK
Risk Permission	Whether account and event controls allow a new position	ALLOWED
Final Action	What the user should do now	NO TRADE YET

Primary deliverables

- Live/near-live XAUUSD dashboard with H4 chart and D1/H1 context.
- BUY, SELL, WAIT FOR CONFIRMATION or NO TRADE output with reasons and risks.
- Entry zone, invalidation/stop, take-profit levels, risk/reward and safe lot size.
- Fundamental dashboard for USD, nominal and real yields, Fed stance, inflation, labor, growth, risk sentiment, flows and event risk.
- Account risk controls, daily loss lock, consecutive-loss cooldown and minimum-lot safety check.
- Trade journal, performance metrics, equity curve and signal history.
- Bias-safe historical backtesting and paper trading before any optional broker execution.

2. Product scope and operating model

2.1 Product objective

Help the user make disciplined H4 XAUUSD decisions while keeping risk proportional to account equity. The daily target (for example USD 10) is displayed as progress only. It must never cause the strategy to raise risk, widen stops, force a trade or use martingale behavior.

2.2 In-scope

Area	Included capability
Market analysis	H4 primary; D1 context; H1 confirmation; structure, trend, momentum, volatility, support/resistance and price action.
Fundamentals	USD, Treasury yields/real yields, Fed policy context, CPI/PCE, labor, GDP/PMI/retail where available, risk sentiment, central-bank/ETF flows as slower inputs.
Event protection	Calendar countdown, importance, pre/post release lock windows and stale-data checks.
Risk	Equity-based risk amount, broker-specific volume calculation, daily loss cap, max trades, consecutive-loss cooldown.
Execution mode	Paper trading by default; live broker execution is a later optional phase with explicit confirmation.
Validation	Backtest, walk-forward/out-of-sample analysis, performance metrics and audit history.

2.3 Explicit non-goals

- Guaranteeing USD 10/day or any fixed return.
- Predicting every H4 candle direction.
- Autonomous live trading in the MVP.
- Using an LLM as a price feed, economic calendar, broker calculation engine or source of numeric facts.
- Martingale, loss-recovery sizing, grid averaging or uncontrolled pyramiding.
- Treating setup score as a win probability unless statistically calibrated out-of-sample.

2.4 User journey

1. User opens the dashboard and selects/loads the trading account profile.
2. System validates XAUUSD data freshness and provider health.
3. Technical and fundamental engines update independently.
4. Economic-event gate determines whether new entries are permitted.
5. Strategy engine detects a setup and proposes entry, stop and targets.
6. Risk engine calculates safe volume from equity and broker specifications.
7. Final decision card shows action, reasoning, risk and invalidation conditions.
8. If paper trading is enabled, user may simulate the trade and the result is journaled automatically.

3. Risk philosophy and account protection

3.1 Risk hierarchy

Priority	Rule
1	Protect capital and obey hard risk limits.
2	Require a technically valid invalidation level and acceptable execution cost.
3	Require event-risk and data-quality permission.
4	Prefer high-quality confluence over trade frequency.
5	Track the daily target only after all safety conditions are satisfied.

3.2 Default account controls

Control	Recommended default	Behavior
Risk per trade	0.50% equity	User may lower it; hard product cap should be configurable, e.g. 1.00%.
Maximum daily realized + open risk	2.00% equity	Block new trades when reached.
Maximum trades/day	3	Prevents signal chasing and overtrading.
Consecutive-loss cooldown	3 losses	New trade signals become non-actionable until cooldown/reset condition.
Minimum planned R:R	2.0 before costs	Reject setups that do not justify risk.
Daily profit target	USD 10 example	Progress-only; never affects lot size.
Martingale / recovery sizing	Disabled	Hard prohibition.

3.3 Hard-stop behavior

When any hard gate is violated, the system returns NO TRADE and the exact reason. Examples: stale data, daily loss cap reached, minimum broker lot exceeds safe size, high-impact event lock, abnormal spread, invalid stop distance or missing symbol specifications.

Safety invariant

If the broker minimum lot would risk more than the configured maximum loss, the only safe output is NO TRADE. Never round position size upward to satisfy the broker minimum.

4. Market data, timekeeping and data quality

4.1 Required market data

Data	Minimum fields	Use
XAUUSD bars	timestamp, open, high, low, close, volume/tick volume	Technical engine and backtest.
Quote	bid, ask, timestamp	Spread and current mark.
Symbol specification	contract size, tick size, tick value, volume min/max/step, currency	Risk and P/L math.
Session metadata	trading hours, holidays, timezone/alignment	Consistent H4 construction and market-open status.

4.2 Timeframe policy

- Store all internal timestamps in UTC.
- Use the market-data provider/broker canonical H4 bar boundaries; do not manufacture local H4 boundaries that disagree with the trading venue/feed.
- Use only closed H4 candles for confirmed signals. A still-forming H4 candle may be displayed but must be marked INTRACANDLE / UNCONFIRMED.
- D1 context and H1 confirmation must come from the same provider or a normalized feed with known session alignment.
- Weekend gaps, maintenance windows and holiday closures must be recognized rather than interpreted as missing data.

4.3 Data quality gates

Check	Fail condition	System response
Freshness	Quote/bar exceeds configured age	DATA STALE; no new signal.
Completeness	Missing expected bars or null OHLC	Quarantine segment; no analysis until repaired.
OHLC validity	low > min(open,close) or high < max(open,close)	Reject malformed bar.
Duplicate timestamp	Multiple canonical bars for same period	Deduplicate deterministically and audit.

Check	Fail condition	System response
Spread	Spread above configured percentile/threshold	NO TRADE or caution state.
Clock drift	Provider timestamp materially ahead/behind server clock	Degrade provider health and block confirmation.

4.4 Provider architecture

```
interface MarketDataProvider {
    getQuote(symbol): Quote
    getCandles(symbol, timeframe, from, to): Candle[]
    getSymbolInfo(symbol): SymbolSpec
    getTradingSession(symbol): SessionInfo
    health(): ProviderHealth
}
```

The platform must support adapters so the trading logic is not tied to one vendor. For real money, broker-provided symbol specifications should be authoritative for volume and P/L calculations.

5. Technical analysis engine

5.1 Multi-timeframe responsibilities

Timeframe	Role	Typical questions
D1	Macro trend context	Is the larger structure trending, ranging or transitioning?
H4	Primary setup timeframe	Where is structure, invalidation, volatility and the trade thesis?
H1	Entry confirmation	Is there a clean trigger/retest without contradicting H4?

5.2 Indicators and features

Feature	Default	Purpose
EMA	20 / 50 / 200	Trend alignment and dynamic context; never standalone.
RSI	14	Momentum state/divergence context; not automatic overbought/oversold reversal.

Feature	Default	Purpose
MACD	12 / 26 / 9	Momentum/crossover confirmation.
ATR	14	Volatility regime, stop buffer, normalization.
ADX	14 (optional but recommended)	Trend-strength/range filter.
Swing structure	Confirmed pivots	HH/HL/LH/LL and break-of-structure logic.
Support/resistance	ATR-normalized swing clusters	Zones, not single magic prices.
Price action	Engulfing, rejection, inside bar, breakout close	Trigger evidence only in context.

5.3 Confirmed swing structure

Use confirmed pivots with a configurable strength, for example two or three bars on each side. In live trading, a pivot becomes known only after the right-side confirmation bars close. The backtester must honor that same delay to avoid look-ahead bias.

Bullish structure: recent confirmed swings show HH + HL progression.
 Bearish structure: recent confirmed swings show LH + LL progression.
 Ranging: directional progression is absent and price repeatedly rotates inside overlapping zones.
 Transition: break of structure occurs but follow-through/confirmation is incomplete.

5.4 Support/resistance zones

- Collect confirmed swing highs/lows over a rolling window.
- Cluster nearby levels using an ATR-normalized distance threshold so the method adapts to volatility.
- Score zones by recency, number of touches/rejections and higher-timeframe overlap.
- Mark the nearest opposing zone and reject entries with insufficient room to target.
- Do not use future pivots when reconstructing historical zones.

5.5 Breakout and retest

A breakout is confirmed only on candle close beyond a zone with a configurable ATR buffer. Wicks alone are not confirmation. The system may optionally require a retest and rejection before entry. Failed breakouts become evidence against immediate continuation.

5.6 Technical setup families

Setup	Core conditions	Typical invalidation
Trend pullback	D1/H4 trend aligned; pullback into EMA/support; H1/H4 rejection; momentum not collapsing	Below/above structural swing or volatility-adjusted zone.
Breakout-retest	H4 close beyond major zone; adequate impulse; retest holds; room to next zone	Close back through breakout zone / retest failure.
Structure reversal	Higher-timeframe exhaustion + H4 structure break + retest + momentum confirmation	Reclaim of prior structure; lower score than trend continuation until validated.

5.7 Technical setup quality score (not probability)

Component	Weight	Notes
H4 market structure	20	Direction and cleanliness of swings.
D1/H4/H1 alignment	15	Multi-timeframe agreement.
EMA / trend regime	10	Alignment and slope, not crossover alone.
Support/resistance location	15	Quality of location and room to target.
Momentum	10	RSI/MACD/ADX context.
Price-action trigger	10	Closed-candle trigger quality.
Volatility / spread regime	5	Avoid dislocated or illiquid conditions.
Entry quality + R:R	15	Distance to invalidation, targets and costs.

Suggested interpretation: 80-100 high-quality setup; 70-79 valid; 60-69 watch; below 60 no trade. These bands must be validated and may be recalibrated. They are not win probabilities.

6. Fundamental analysis engine

Fundamental analysis should be maintained as a separate evidence layer. World Gold Council research emphasizes that gold is influenced by interacting drivers such as risk/uncertainty, opportunity cost (rates and currencies), economic expansion and momentum. The app therefore must avoid simplistic one-factor rules. [S1]

6.1 Fundamental driver map

Driver	Data / proxy	Typical gold interpretation	Important caveat
US dollar	Licensed DXY/FX data	A weaker USD often supports USD-priced gold	Correlation is regime-dependent.
Nominal yields	US 2Y/10Y Treasury	Falling yields can reduce opportunity cost	Growth/risk shocks can complicate response.
Real yields	TIPS real-yield curve	Lower real yields can support gold	Since 2022, other drivers have at times offset the classic inverse relation. [S2]
Fed policy / expectations	FOMC statement, projections, market expectations	Dovish repricing can support gold; hawkish repricing can pressure it	Use actual market reaction, not text sentiment alone.
Inflation	CPI, PCE and core measures	Affects real rates and Fed path	High inflation can strengthen USD/yields and hurt gold initially.
Labor	NFP, unemployment, wages, claims	Changes expected Fed path and growth outlook	Headline NFP alone is insufficient.
Growth	GDP, PMI/ISM, retail sales	Slowdown can affect yields and safe-haven demand	Interpret jointly with inflation and policy.
Risk / geopolitics	Market volatility + reputable news	Can raise safe-haven demand	Do not sensationalize; weight modestly.
Flows	Gold ETF flows / futures positioning	Shows investor demand and momentum	Slower-moving; not an entry trigger by itself.
Central banks	Official/WGC/IMF-style reserve data	Structural demand context	Medium/long horizon, not intraday trigger.

6.2 Source hierarchy

Tier	Preferred source	Use
1 - Primary official	Federal Reserve, BLS, BEA, U.S. Treasury	Policy, inflation, labor, GDP/PCE, yields.
2 - Licensed market data	Broker or reputable market-data vendor	XAUUSD, USD index/FX, real-time rates, consensus calendar.
3 - Specialist research	World Gold Council and comparable research	Gold-specific flows and structural context.
4 - News/sentiment	Reputable structured news feed	Event awareness and narrative; never replace hard data.

Tier	Preferred source	Use
Never	LLM memory or invented values	Not a data source.

6.3 Event surprise normalization

Do not combine raw percentage-point differences across indicators. Where consensus data is licensed, standardize surprises using each indicator's historical forecast-error distribution and a direction map.

```
raw_surprise = actual - consensus
standardized_surprise = raw_surprise / rolling_std(forecast_error)

driver_score = clamp(standardized_surprise * direction_weight, -1, +1)
# Revisions and simultaneous subcomponents (e.g., wages + unemployment + payrolls)
# must be included before translating the release into a gold-impact score.
```

6.4 Fundamental score

Component	Illustrative max weight
USD / broad dollar direction	20
Nominal + real yields	20
Fed policy / rate expectations	20
Inflation composite	15
Labor composite	10
Risk/geopolitical regime	5
Flows / central-bank context	5
Growth / secondary macro	5

Each component produces a signed gold-impact score. The aggregate may be shown from -100 to +100: negative = bearish gold context, positive = bullish gold context. Weights are configuration values and must be validated rather than assumed optimal.

6.5 Fundamental output

```
Fundamental Bias: BULLISH
Score: +54
Main supporting drivers:
- USD weakening on the selected horizon
- 10Y real yield falling
- Rate expectations becoming less restrictive
```

Conflicting evidence:

- Inflation surprise remains elevated

Upcoming risk:

- High-impact CPI release inside protection window

7. Economic calendar and event-risk controls

7.1 Event classes

Impact	Examples	Default behavior
Very high	FOMC decision/press conference, CPI, NFP	Block new entries before release and for a configurable stabilization period after.
High	PCE, GDP, major Fed speech, ISM, retail sales	Block or require elevated threshold depending on historical XAUUSD response.
Medium	Claims, secondary surveys/data	Caution state; may reduce setup quality.
Low	Minor releases	Informational only unless spread/volatility changes materially.

7.2 Default event windows

Use configurable windows rather than hard-coded universal values. A conservative starting point is 30 minutes before and 30 minutes after major scheduled releases, with wider windows for FOMC/NFP/CPI if desired. The post-event lock should be extendable until spread and realized volatility normalize.

7.3 News-risk states

- LOW - no major scheduled risk nearby.
- ELEVATED - important event approaching or volatility rising.
- HIGH - inside protection window; no new actionable entry.
- POST-EVENT STABILIZATION - wait until quote quality/spread/volatility normalize.

8. Confluence and final decision engine

8.1 Gating order

9. Data gate: market data, symbol specs and account data are valid and fresh.
10. Risk gate: account is not locked by daily loss, trade count or cooldown.
11. Event gate: not inside a restricted macro/news window.
12. Market-regime gate: volatility/spread and structure support the setup family.
13. Directional evidence: technical and fundamental biases are calculated independently.

14. Setup gate: confirmed entry logic, stop and targets exist.
15. Economics gate: planned R:R remains acceptable after expected costs.
16. Sizing gate: safe lot size is at least the broker minimum without exceeding maximum risk.
17. Final state: BUY, SELL, WAIT FOR CONFIRMATION or NO TRADE.

8.2 Decision matrix

Technical	Fundamental	Event risk	Example action
Bullish	Bullish	Low	BUY only if a valid entry trigger and risk gates pass.
Bullish	Neutral	Low	Possible BUY with normal/reduced quality depending on location.
Bullish	Bearish	Low	WAIT or NO TRADE unless technical setup is exceptionally strong and policy permits.
Bearish	Bearish	Low	SELL only if a valid trigger and risk gates pass.
Any	Any	High	NO TRADE / event lock.
Any	Any	Low but unsafe size	NO TRADE / account-size constraint.

8.3 Explainability contract

Every final state must carry structured reasons, not free-form guesses: supporting evidence, conflicting evidence, invalidation condition, event risk, data quality, account risk and the exact gate that blocked a trade if applicable.

9. Position sizing and trade math

9.1 Core equations

```

risk_amount = equity * risk_percent
stop_distance = abs(entry_price - stop_price)
stop_ticks = stop_distance / tick_size
loss_per_lot_before_costs = stop_ticks * tick_value_per_lot
estimated_loss_per_lot = loss_per_lot_before_costs + cost_buffer_per_lot
raw_volume = risk_amount / estimated_loss_per_lot
safe_volume = floor(raw_volume / volume_step) * volume_step

```

Use broker-provided tick size, tick value, contract size and volume rules. Do not assume that every XAUUSD broker uses the same contract specification. The implementation should use bid/ask-aware calculations so spread is not double-counted.

9.2 Minimum lot rule

```
if safe_volume < volume_min:
    risk_at_min = estimate_loss(volume_min, entry, stop, costs)
    if risk_at_min > risk_amount:
        return NO_TRADE("Broker minimum volume exceeds configured risk")
```

9.3 Illustrative example (not live market data)

Input	Illustrative value
Equity	USD 1,000
Risk per trade	0.50%
Maximum planned loss	USD 5.00
Entry	4,000.00
Stop	3,995.00
Tick size	0.01
Tick value at 1.00 lot	USD 1.00 per tick (example only)
Stop ticks	500
Loss per 1.00 lot before costs	USD 500
Raw volume	0.010 lot before cost buffer

With real trading, the cost buffer (spread, commission and slippage assumption) can reduce the safe volume. If the rounded-down volume falls below the broker minimum, the system must reject the trade.

9.4 Stop and target policy

- Stops are placed at technical invalidation, with optional ATR buffer; never choose a stop merely to manufacture a desired lot size.
- Targets are based on opposing structure/liquidity zones and validated against minimum net R:R.
- If a natural stop is too wide for the account at minimum lot, the trade is not affordable and must be skipped.
- Partial take-profit rules, break-even moves and trailing stops should be separate strategy variants and backtested independently rather than mixed ad hoc.

10. Trade lifecycle and paper trading

10.1 Setup states

State	Meaning
WATCHING	Bias exists but entry criteria are incomplete.
ARMED	Conditions close to valid; waiting for trigger/close.
VALID	All gates pass and a trade plan is actionable.
BLOCKED	Hard risk/event/data gate prevents action.
EXPIRED	Setup invalidated or entry window passed.
OPEN	Paper/live position exists.
CLOSED	Trade outcome finalized and journaled.

10.2 Paper trading requirements

- Paper mode is the default for all new users/environments.
- Simulate bid/ask, spread, commission and slippage assumptions rather than filling at impossible mid prices.
- Persist the exact strategy version, indicator snapshot, fundamental snapshot and risk settings used at entry.
- Do not rewrite historical signal inputs after outcomes are known.
- Allow manual cancellation but preserve audit history.

10.3 Daily reset and timezone

Trading-day boundaries should be configurable, but risk ledgers should use a clearly defined timezone and exchange/broker session convention. Do not mix local-browser date boundaries with broker-session risk accounting without an explicit rule.

11. Backtesting and quantitative validation

11.1 Non-negotiable bias controls

Risk	Required control
Look-ahead bias	Only information known at the historical timestamp may be used. Confirmed pivots become available only after their confirmation bars.
Same-bar ambiguity	If both SL and TP could be hit inside one OHLC bar, use lower-timeframe reconstruction or a conservative rule; do not

Risk	Required control
	always choose the favorable outcome.
Survivorship/data revision	Store vintage macro data where possible; revisions must not silently replace what was known at the time.
Cost omission	Model spread, commission and slippage.
Overfitting	Separate development, validation and untouched out-of-sample periods; use walk-forward evaluation.
Parameter mining	Track all variants tested; do not select only the best result without multiple-testing awareness.

11.2 Backtest outputs

Metric	Why it matters
Net P/L and return	Overall result, but not sufficient alone.
Expectancy in R	Average value per trade normalized by risk.
Profit factor	Gross profit / gross loss.
Max drawdown	Capital stress and risk-of-ruin context.
Win rate	Useful only alongside payoff ratio.
Average/median R	Distribution of trade outcomes.
Consecutive losses	Supports cooldown and capital planning.
Exposure/time in market	Measures how selective the system is.
Sharpe / Sortino (optional)	Risk-adjusted context, with care for non-normal returns.
Regime breakdown	Trend/range, high/low volatility, event vs non-event periods.

11.3 Validation sequence

18. Unit-test calculations.
19. Replay historical candles with deterministic outputs.
20. Run an in-sample development period.
21. Freeze rules and evaluate untouched out-of-sample data.
22. Perform walk-forward analysis across multiple market regimes.
23. Stress costs and slippage.
24. Run paper trading forward in real time.

25. Only after stable results, consider a tightly controlled live pilot.

12. Performance analytics

12.1 Daily dashboard metrics

- Realized P/L and open risk.
- Daily target progress (informational).
- Risk used vs remaining risk budget.
- Trades taken vs daily maximum.
- Wins, losses, average R and current loss streak.
- Signal count vs executed paper trades.

12.2 Long-horizon metrics

- Equity curve and underwater drawdown chart.
- Monthly return and rolling expectancy.
- Profit factor, win rate, average win/loss and payoff ratio.
- Performance by setup family and market regime.
- Performance by technical/fundamental alignment state.
- Performance around event windows vs normal periods.
- Calibration of setup-quality buckets against realized outcomes.

12.3 Expectancy

```
expectancy = (win_rate * average_win) - (loss_rate * average_loss)
# Prefer expressing average_win and average_loss in R-multiples.
# Positive historical expectancy does not guarantee future returns.
```

13. UI/UX specification

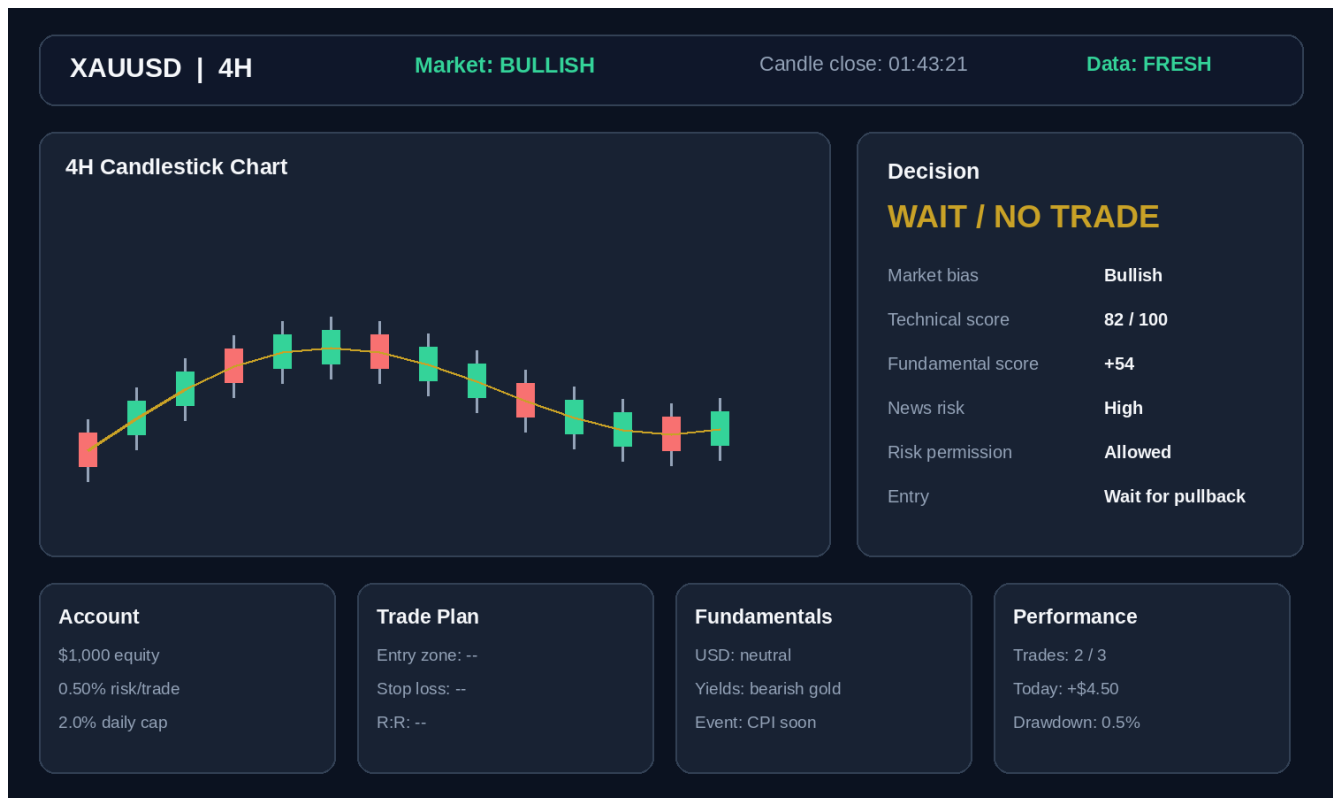


Figure 2. Conceptual desktop dashboard - final visual design can change, but information hierarchy should remain.

13.1 Top-level dashboard

Area	Required content
Header	XAUUSD bid/ask, spread, market state, H4 candle countdown, data freshness, next high-impact event.
Chart	Candlesticks; EMA20/50/200; optional support/resistance, swing points, entry/SL/TP overlays.
Decision card	Final action, setup quality, market bias, fundamental bias, event risk, entry, stop, targets, R:R, safe lot and dollar risk.
Account card	Equity, configured risk %, daily risk used, trade count, P/L target progress, lock status.
Fundamental panel	USD, yields/real yields, Fed/macro summary, risk regime and event countdown.
Explanation	Supporting reasons, conflicting evidence, invalidation and exact block reason if no trade.

13.2 Critical UX rules

- The first screen must answer: what is gold doing, what is the action, why, where is invalidation, how much is at risk and what could change the view?
- Color must reinforce text, never replace it. BUY/SELL/NO TRADE states require explicit labels.
- Show data timestamp and provider health near every time-sensitive panel.
- If data is stale, hide actionable entry controls and show a blocking banner.
- If the user changes equity or risk %, immediately recompute risk and volume without changing the strategy signal itself.

14. System architecture

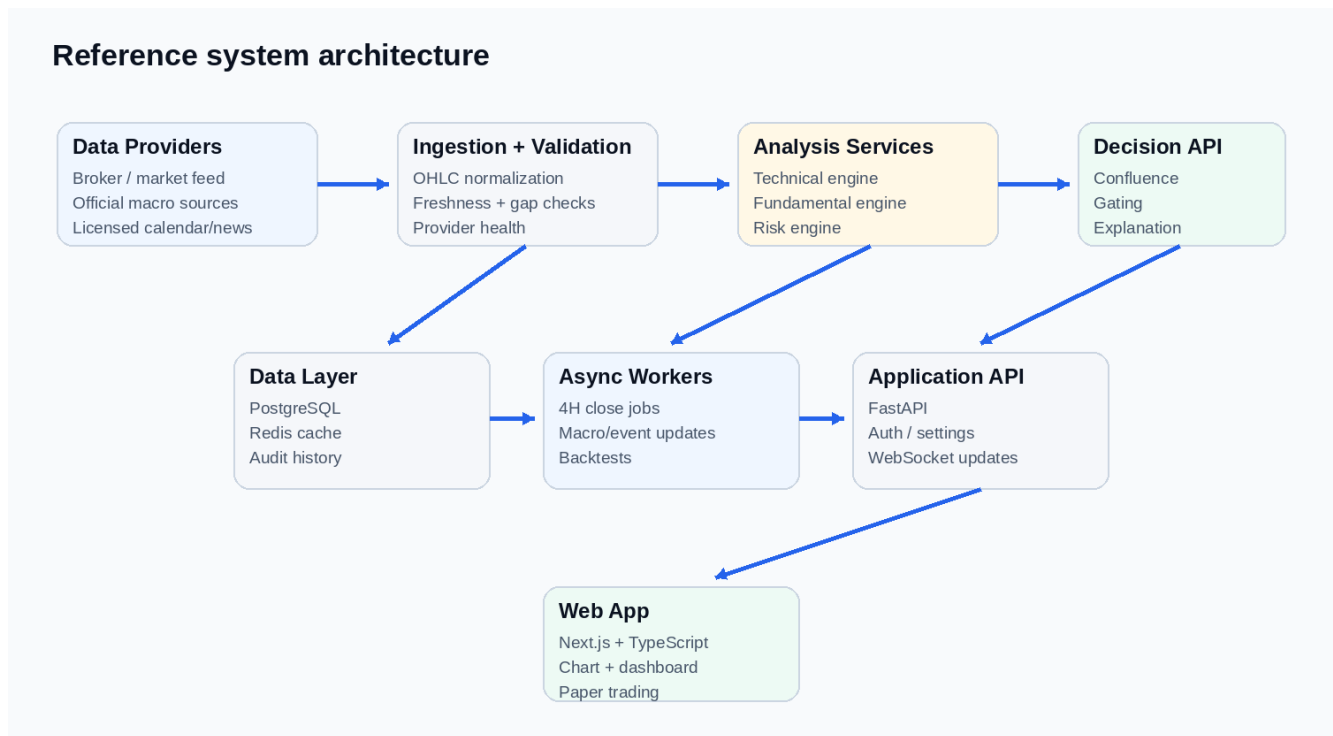


Figure 3. Recommended modular architecture; providers and execution adapters remain replaceable.

14.1 Recommended stack

Layer	Recommendation
Web UI	Next.js + TypeScript + Tailwind CSS + shadcn/ui.
Charting	TradingView Lightweight Charts or equivalent embeddable financial chart.
API / analytics	FastAPI (Python) for market/strategy/risk services.
Quant libraries	pandas, numpy; indicators implemented or verified against a trusted library.
Database	PostgreSQL; TimescaleDB extension optional if time-series

Layer	Recommendation
	scale warrants it.
Cache / queue	Redis; worker queue for scheduled analysis and backtests.
Auth	Secure session/OAuth provider depending deployment context.
Deployment	Docker containers; managed Postgres recommended for production.

14.2 Service boundaries

- Market Data Service - provider adapters, normalization, sessions and symbol specs.
- Technical Engine - deterministic indicator/structure features and setup detection.
- Fundamental Engine - macro observations, event surprises, source provenance and driver score.
- Risk Engine - account limits, position sizing, cost model and permission state.
- Decision Engine - gating and confluence; produces structured final decision.
- Paper Trading Engine - simulated orders, fills, positions and journal.
- Backtest Worker - historical replay using versioned strategies.
- Explanation Layer - converts structured reasons into user-readable narrative; LLM optional.

15. Data model

15.1 Core tables

Table	Key fields / purpose
users	identity, timezone, preferences
trading_accounts	user_id, name, currency, equity snapshot, mode
broker_symbol_specs	provider, symbol, contract_size, tick_size, tick_value, volume_min/max/step, effective_at
risk_profiles	risk_per_trade, daily_loss_cap, max_trades, loss_streak_limit, min_rr
market_bars	provider, symbol, timeframe, ts_open, OHLCV, quality flags
market_quotes	bid, ask, timestamp, provider, freshness
indicator_snapshots	analysis_run_id, EMA/RSI/MACD/ATR/ADX and derived features
fundamental_observations	source, series/event, actual, prior, consensus, revision, timestamp, provenance

Table	Key fields / purpose
economic_events	event_id, scheduled_at, importance, status, actual/consensus/prior
analysis_runs	strategy_version, market snapshot, technical score, fundamental score, event state
trade_setups	direction, state, entry zone, stop, targets, score, expiry, reasons
paper_orders	order type, volume, requested/filled price, cost assumptions, state
positions	average entry, volume, stop, targets, unrealized P/L
trades	open/close timestamps, realized P/L, R-multiple, exit reason
daily_risk_ledger	date/session, starting equity, realized P/L, open risk, trades count, lock state
backtest_runs	strategy version, dataset, parameters, costs, results summary
backtest_trades	run_id, setup snapshot, entry/exit, P/L, R
provider_health	provider, latency, last_success, stale/error flags
audit_log	actor, action, before/after, timestamp

15.2 Data lineage requirement

Every analysis result must be reproducible. Store provider/source identifiers, timestamps, strategy version, configuration hash and the specific market/fundamental snapshot used. Never depend solely on mutable latest-value tables for historical reconstruction.

16. API design

16.1 Primary endpoints

Method / endpoint	Purpose
GET /api/market/xauusd/quote	Current bid/ask, spread and freshness.
GET /api/market/xauusd/candles?tf=4h	Normalized candles.
GET /api/fundamentals/xauusd	Current fundamental driver snapshot.
GET /api/calendar	Upcoming economic events and protection state.
POST /api/analysis/xauusd/run	Create a deterministic analysis run.

Method / endpoint	Purpose
GET /api/analysis/xauusd/latest	Latest final decision and explanation inputs.
POST /api/risk/quote	Calculate safe volume and expected loss/target P/L.
POST /api/paper/orders	Create a paper order from a valid setup.
GET /api/performance	Account and strategy metrics.
POST /api/backtests	Create an asynchronous backtest run.
GET /api/backtests/{id}	Status and result summary.
PUT /api/settings/risk	Update user risk controls.

16.2 Decision response example

```
{
  "symbol": "XAUUSD",
  "timeframe": "H4",
  "marketBias": "BULLISH",
  "tradeSetup": "WAITING_FOR_PULLBACK",
  "riskPermission": "ALLOWED",
  "finalAction": "NO_TRADE",
  "technicalScore": 82,
  "fundamentalScore": 54,
  "newsRisk": "HIGH",
  "entry": null,
  "stopLoss": null,
  "targets": [],
  "safeVolume": null,
  "blockReasons": ["High-impact event inside protection window"],
  "dataQuality": "FRESH",
  "strategyVersion": "h4-gold-v1.0"
}
```

17. Security, reliability and observability

17.1 Security controls

- Keep provider/broker API keys server-side in a secret manager; never expose them to the browser.
- Use TLS, secure cookies/session handling, CSRF protection where applicable and least-privilege database roles.
- Encrypt sensitive account/configuration data at rest where appropriate.
- Add rate limiting to analysis and backtest endpoints.
- Separate paper and live credentials/environments; live execution requires explicit enablement and confirmation.
- Audit settings changes, risk overrides and every order action.

17.2 Reliability controls

- Provider health scoring and automatic degradation to non-actionable state on stale/failed feeds.
- Idempotent H4-close analysis jobs so duplicate schedules do not create duplicate setups.
- Database uniqueness constraints on provider+symbol+timeframe+bar timestamp.
- Retries with bounded backoff for transient data-provider failures.
- No silent fallback to fabricated/default market values.

17.3 Observability

Signal	Examples
Metrics	data latency, job duration, analysis errors, stale-data incidents, API latency, backtest queue depth
Logs	provider calls, normalized timestamps, gating outcomes, risk calculations, paper fills
Alerts	data stale, provider down, job missed, database error, repeated calculation mismatch
Audit	strategy version/config hash, manual overrides, settings changes, execution confirmations

18. Testing strategy

18.1 Unit tests

- EMA/RSI/MACD/ATR/ADX numerical correctness.
- Pivot confirmation and structure classification.
- Support/resistance clustering.
- Event-window calculation and timezone/DST behavior.
- Risk amount, tick conversion, cost model, lot rounding and minimum-lot rejection.
- Daily risk ledger and cooldown transitions.
- Score aggregation and deterministic decision gates.

18.2 Integration tests

- Market provider -> normalized bars -> technical analysis.
- Macro source -> normalized observation -> fundamental score.
- Economic calendar -> event lock -> final decision.
- Risk settings -> setup -> safe volume -> paper order.
- Provider outage/stale data -> actionable controls disabled.

18.3 Historical/replay tests

- Exact same historical input and strategy version must reproduce the same decision.
- No post-hoc use of revised macro data unless the test explicitly models revised data.
- Same-bar stop/target ambiguity handled conservatively.
- H4 signal time equals closed-bar availability time, not candle open time.

18.4 Adversarial edge cases

Case	Expected result
Equity = 0 / negative	Validation error; no sizing.
Stop distance = 0	Reject setup.
Tick value missing	Block sizing; request broker spec.
Min lot too risky	NO TRADE.
Spread spikes	NO TRADE / caution.
Both TP and SL hit same historical bar	Conservative/low-timeframe resolution.
High-impact event starts soon	Entry blocked.
Provider returns future timestamp	Provider degraded and analysis blocked.

19. Deployment and operations

19.1 Environments

Environment	Purpose
Development	Mock feeds and unit/integration tests.
Staging	Live read-only feeds, paper trading, production-like scheduling.
Production - paper	Real-time decision support and paper fills.
Production - live (optional)	Only after explicit readiness review; manual-confirmation execution first.

19.2 Container layout

```

web      -> Next.js UI
api      -> FastAPI application
worker   -> scheduled H4/macro/backtest jobs
postgres -> persistent data
redis    -> cache + queue
reverse-proxy / managed ingress -> TLS and routing

```

19.3 Backup and retention

- Daily database backups with tested restore procedure.
- Retain raw/normalized market data according to provider licensing.
- Preserve signal/trade/audit history needed for reproducibility.
- Rotate application logs separately from trading audit records.

20. Development roadmap

Phase	Deliverables	Exit criteria
0 - Guardrails	Requirements, symbol-spec model, risk policy, provider contracts	Risk invariants approved; paper-only MVP confirmed.
1 - Foundation	Repo, Docker, DB migrations, auth, settings, mock dashboard	Tests/lint/type checks pass.
2 - Market data	Provider adapter, bars/quotes/specs, H4 countdown, data quality	Live read-only feed stable; stale-data blocking works.
3 - Technical engine	Indicators, structure, zones, setup families, technical score	Historical replay deterministic; no look-ahead tests pass.
4 - Risk engine	Sizing, daily limits, cost model, permission state	Minimum-lot and daily-lock tests pass.
5 - Fundamental engine	Official macro ingestion, yields, event model, source lineage	No invented data; timestamps/provenance visible.
6 - Decision engine	Confluence, gating, structured explanations	BUY/SELL/WAIT/NO TRADE behavior auditable.
7 - Paper trading	Orders, fills, positions, journal, daily P/L	Paper lifecycle reliable with costs.
8 - Backtesting	Bias-safe replay, metrics, walk-forward support	Out-of-sample report generated and reviewed.
9 - UX hardening	Responsive dashboard, errors, accessibility, settings	Critical information readable in <5 seconds.
10 - Production hardening	Monitoring, alerts, backups, security review	Paper-production readiness checklist passes.
11 - Optional live pilot	Broker execution adapter + explicit confirmation	Separate formal approval; small controlled exposure.

21. Acceptance criteria

21.1 MVP acceptance

- User can input equity and risk settings and receive a mathematically correct safe-volume result from broker symbol specs.

- Dashboard shows fresh XAUUSD H4 data, D1/H1 context and exact data timestamp.
- Technical engine produces deterministic market regime, score and structured reasons.
- Fundamental engine shows sourced macro/yield/event context with timestamps and provenance.
- Decision engine can return BUY, SELL, WAIT FOR CONFIRMATION or NO TRADE and identifies every blocking gate.
- Paper trading records entry, exit, costs, P/L and R-multiple.
- Daily risk limits and minimum-lot rejection cannot be bypassed by UI behavior.
- Backtest engine demonstrates no obvious look-ahead leakage and produces core metrics.
- Automated tests, linting and type checking pass in CI.

21.2 Live-readiness gate (future)

Do not skip

Before any real-money execution: sustained paper-forward results, independent review of risk calculations, broker-specific fill/cost testing, credential/security review, operational monitoring, jurisdiction/legal review and explicit user confirmation on every order in the first live phase.

22. Recommended initial defaults

Setting	Starting value	Rationale
Primary timeframe	H4	Matches requested operating horizon.
Context timeframes	D1 + H1	Higher-level bias plus lower confirmation.
Risk/trade	0.50%	Conservative starting point; configurable downward.
Daily loss cap	2.00%	Hard capital-protection limit.
Max trades/day	3	Discourages overtrading.
Loss-streak cooldown	3	Pause after repeated failure.
Minimum net R:R	2.0	Filters low-payoff trades; validate historically.
Technical action threshold	≥ 70 valid, ≥ 80 high quality	Setup-quality labels only; not probability.
High-impact pre-event lock	30 min starting point	Configurable by event type and empirical response.
High-impact post-event lock	30 min + normalization gate	Extend if spread/volatility remains abnormal.
Paper mode	ON	Default until validation is complete.

These values are engineering defaults, not claims of optimality. Strategy thresholds and event windows should be validated against historical and forward paper data.

23. Example decision scenarios

Scenario A - Technical and fundamental alignment, valid risk

Layer	State
Market bias	BULLISH
Technical	82/100; H4 trend pullback at support
Fundamental	+48; supportive but not extreme
Event risk	LOW
Risk permission	ALLOWED
Entry economics	Net R:R 2.3 after estimated costs
Result	BUY setup may be actionable after confirmed trigger

Scenario B - Good chart, high event risk

Layer	State
Market bias	BULLISH
Technical	84/100
Fundamental	Neutral / mixed
Event risk	HIGH - CPI release inside protection window
Risk permission	Account allowed
Result	NO TRADE until event lock clears and spread/volatility normalize

Scenario C - Valid setup but account too small for broker minimum

Layer	State
Equity	USD 100

Layer	State
Configured risk	0.50% = USD 0.50
Technical setup	Valid
Broker minimum	0.01 lot
Estimated loss at minimum lot	USD 4.00
Result	NO TRADE - minimum volume exceeds configured maximum risk

Scenario D - Daily target not reached

Layer	State
Daily target	USD 10
Realized today	USD 2
Market	Range / conflicting technical evidence
Fundamental	Mixed
Result	NO TRADE - system does not increase risk or manufacture a setup to chase the remaining USD 8

Appendix A. Master AI build prompt

Copy the following prompt into your coding AI. It is intentionally explicit so the implementation agent treats this document as an engineering specification rather than a request for a simple signal page.

ROLE

You are a senior quantitative trading engineer, financial data engineer, backend engineer, frontend engineer and product-quality reviewer.

MISSION

Build a production-quality web application for XAUUSD decision support on the H4 timeframe. The application must be risk-first, explainable and paper-trading-first. It must never promise profit and must never force a trade because a daily target has not been reached.

NON-NEGOTIABLE OUTPUT MODEL

Return four independent states:

1. Market Bias: STRONG_BULLISH / BULLISH / NEUTRAL / BEARISH / STRONG_BEARISH
2. Trade Setup: WATCHING / ARMED / VALID / BLOCKED / EXPIRED
3. Risk Permission: ALLOWED / BLOCKED
4. Final Action: BUY / SELL / WAIT_FOR_CONFIRMATION / NO_TRADE

PRIMARY TIMEFRAMES

- D1 = higher-timeframe context
- H4 = primary signal timeframe
- H1 = entry confirmation

Confirmed H4 signals MUST use closed candles. Intrabar observations must be marked UNCONFIRMED.

DATA ARCHITECTURE

Create replaceable provider interfaces for:

- XAUUSD quotes and OHLCV
- broker symbol specifications
- economic calendar / consensus
- USD / DXY or licensed equivalent
- Treasury nominal and real yields
- official macro data
- structured news/risk context

Never use an LLM as a numeric data source.

TECHNICAL ENGINE

Implement and test:

- EMA 20 / 50 / 200
- RSI 14
- MACD 12 / 26 / 9
- ATR 14
- ADX 14 optional/recommended
- confirmed swing pivots and HH/HL/LH/LL structure
- ATR-normalized support/resistance zones
- breakout close + optional retest
- price-action trigger context
- trend pullback, breakout-retest and controlled reversal setup families
- D1/H4/H1 alignment

Produce a 0-100 SETUP QUALITY SCORE, never call it a win probability.

FUNDAMENTAL ENGINE

Keep it separate from technical analysis. Score gold impact from -100 to +100 using configurable, validated weights for:

- broad USD direction
- nominal and real yields
- Fed policy / rate expectations
- CPI/PCE inflation
- NFP, unemployment, wages, claims
- GDP and secondary growth indicators

- risk/geopolitical regime
- ETF/futures flows and central-bank demand as slower context

Relationships are context-dependent. Do not encode 'USD up = gold down' as an absolute law.
Use official primary data where available and licensed market data for real-time/consensus inputs.

EVENT RISK

Create LOW / ELEVATED / HIGH / POST_EVENT_STABILIZATION states.
Support configurable pre/post windows by event type. While HIGH, block new actionable entries.
After a release, keep blocking if spread/realized volatility remains abnormal.

RISK ENGINE

User inputs equity and a risk percentage. Calculate:
 $\text{risk_amount} = \text{equity} * \text{risk_percent}$
 $\text{stop_ticks} = \text{abs}(\text{entry} - \text{stop}) / \text{tick_size}$
 $\text{estimated_loss_per_lot} = \text{stop_ticks} * \text{tick_value_per_lot} + \text{modeled_costs_per_lot}$
 $\text{raw_volume} = \text{risk_amount} / \text{estimated_loss_per_lot}$
 $\text{safe_volume} = \text{floor}(\text{raw_volume} / \text{volume_step}) * \text{volume_step}$
 Use broker-specific contract/tick/volume specifications. Never hard-code universal XAUUSD pip values.
 If $\text{safe_volume} < \text{volume_min}$ and the broker minimum would exceed the risk amount, return NO_TRADE.

DEFAULT CAPITAL CONTROLS

- risk/trade: 0.50% starting default
- daily loss cap: 2.00%
- max trades/day: 3
- consecutive-loss cooldown: 3
- minimum net R:R: 2.0
- martingale/recovery sizing: prohibited

The daily USD profit target is display-only and MUST NOT alter risk or volume.

STOP/TARGET RULES

Stops come from technical invalidation plus an optional volatility buffer, not from desired lot size.
Targets come from opposing structure/liquidity zones. Reject trades with insufficient net R:R after costs.
Do not improvise partial exits, break-even or trailing logic; model each as a versioned strategy variant.

DECISION PIPELINE

Data quality -> account risk -> event risk -> regime -> technical bias -> fundamental bias -> setup trigger -> stop/targets -> R:R -> safe volume -> final decision.
Every blocked decision must return exact machine-readable reasons.

PAPER TRADING

Paper mode is the default. Simulate bid/ask, spread, commission and slippage.
Persist the exact strategy version and all market/fundamental/risk inputs at entry.
Never rewrite historical signals after results are known.

BACKTESTING

Implement historical replay with no look-ahead bias. Confirmed pivots become known only after their right-side confirmation bars. If both SL and TP could hit inside one OHLC bar, reconstruct with lower timeframe data or use a conservative rule. Model transaction costs. Support train/validation/out-of-sample separation and walk-forward evaluation.

PERFORMANCE

Calculate net P/L, R-multiple distribution, expectancy, profit factor, max drawdown, win rate, average win/loss, consecutive losses, exposure, regime breakdown and setup-family performance. Setup score may be calibrated later only with out-of-sample evidence.

UI

Professional dark dashboard. Above the fold show:

- XAUUSD bid/ask and spread
- H4 candle countdown
- data freshness
- Market Bias

- Trade Setup
- Risk Permission
- Final Action
- technical score
- fundamental score
- event risk
- entry / SL / TP / net R:R
- safe volume and dollar risk
- equity, daily risk used, trade count and target progress
- next high-impact event

The chart should support candlesticks, EMA overlays, structure zones and trade levels.

STACK

Frontend: Next.js + TypeScript + Tailwind + shadcn/ui

Chart: TradingView Lightweight Charts or equivalent

Backend/analytics: FastAPI + Python

Data: PostgreSQL; Redis for cache/queue

Deployment: Docker

Keep strategy/risk calculations out of UI components.

DATABASE

At minimum implement:

users, trading_accounts, broker_symbol_specs, risk_profiles, market_bars, market_quotes, indicator_snapshots, fundamental_observations, economic_events, analysis_runs, trade_setups, paper_orders, positions, trades, daily_risk_ledger, backtest_runs, backtest_trades, provider_health, audit_log.

Store source/provenance, strategy version and configuration hash so decisions are reproducible.

SECURITY/RELIABILITY

Keep secrets server-side. Add authentication, audit logs, rate limiting, provider health, stale-data blocking, idempotent scheduled jobs, database constraints and explicit paper/live separation.

Never silently substitute fabricated prices or macro values.

TESTS

Write unit tests for indicators, pivots, scores, event windows, risk math, volume rounding, daily locks,

and edge cases. Write integration tests for provider -> analysis -> risk -> paper trade. Add historical

replay tests proving deterministic behavior and no obvious look-ahead leakage.

BUILD PHASES

0 guardrails and requirements

1 project foundation + DB + mock UI

2 market-data integration + quality gates

3 technical engine

4 risk engine

5 fundamental/event engine

6 confluence/final decision engine

7 paper trading + journal

8 backtesting + performance analytics

9 UI hardening + accessibility

10 monitoring/security/deployment hardening

11 optional live broker integration only after explicit readiness review

WORKING METHOD

For every phase:

1. design interfaces and migrations first

2. implement in small modules

3. add automated tests

4. run lint + type checking + tests

5. run the application

6. fix all errors before continuing

Do not create one giant file. Do not skip tests. Do not fake live data.

FIRST IMPLEMENTATION TASK

Create the monorepo/project structure, Docker setup, PostgreSQL schema, provider interfaces, TypeScript/Python domain models, a mock-data dashboard, deterministic risk calculator and unit tests. The first UI must already support the four-state output model and show why a trade is blocked. Only after this foundation passes tests should you connect a live read-only XAUUSD data provider.

Appendix B. Formula reference

Formula	Definition
Risk amount	equity x risk_percent
Stop distance	abs(entry - stop)
Stop ticks	stop_distance / tick_size
Loss per lot	stop_ticks x tick_value_per_lot + modeled transaction-cost buffer
Raw volume	risk_amount / loss_per_lot
Safe volume	floor(raw_volume / volume_step) x volume_step
Planned R:R	expected net reward / expected net risk
R-multiple realized	realized P/L / initial planned risk amount
Profit factor	gross profit / gross loss
Expectancy	win_rate x avg_win - loss_rate x avg_loss

Cost-model note

Use either explicit bid/ask simulation or a documented cost buffer, and ensure spread is counted exactly once. Commission may be per side or round turn depending on broker. Slippage assumptions should be stress-tested, especially around high-impact events.

Appendix C. Research sources

The following sources support the fundamental-data hierarchy, gold-driver framework and risk warnings. Retrieval date: 4 October 2026.

[S1] World Gold Council - Gold Mid-Year Outlook 2026: Point break
<https://www.gold.org/goldhub/research/gold-mid-year-outlook-2026>

[S2] World Gold Council - Are fiscal concerns driving gold? (discussion of real-rate relationship and offsetting drivers)
<https://www.gold.org/goldhub/gold-focus/2025/06/you-asked-we-answered-are-fiscal-concerns-driving-gold>

[S3] U.S. Department of the Treasury - Interest Rate Statistics / Treasury yield curves
<https://home.treasury.gov/policy-issues/financing-the-government/interest-rate-statistics/>

[S4] U.S. Bureau of Labor Statistics - Consumer Price Index
<https://www.bls.gov/cpi/news.htm>

[S5] U.S. Bureau of Labor Statistics - Employment Situation
<https://www.bls.gov/bls/news-release/empst.htm>

[S6] U.S. Bureau of Economic Analysis - Personal Consumption Expenditures Price Index
<https://www.bea.gov/data/personal-consumption-expenditures-price-index>

[S7] U.S. Bureau of Economic Analysis - Gross Domestic Product
<https://www.bea.gov/data/gdp/gross-domestic-product>

[S8] Federal Reserve - FOMC meeting calendars, statements and minutes
<https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>

[S9] CFTC - Eight Things You Should Know Before Trading Forex
https://www.cftc.gov/LearnAndProtect/AdvisoriesAndArticles/CustomAdvisory_MustKnowForex.html

Accuracy notes

- Official macroeconomic releases can be revised. Historical analysis should preserve the vintage available at the decision timestamp where feasible.
- Consensus forecasts and real-time DXY/market prices generally require a licensed commercial provider; official agencies publish actual releases but not universal market consensus feeds.
- Broker XAUUSD contract specifications, sessions, tick values, margin and execution costs can differ. The application must retrieve or configure them per broker/account.
- The relationships described in this document are analytical drivers, not guaranteed causal rules or return forecasts.

Final implementation principle

The best version of this product is not the one that produces the most BUY/SELL signals. It is the one that reliably says NO TRADE when data, setup quality, event risk, execution cost or account safety does not justify exposure.